

Test 6 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.NSO.3.3 moderate	5^2 , 5×5 , 25×1	A power repeats the base as a factor; it is not the base times the exponent.
2	MA.6.DP.1.4 moderate	0, 3	Look for the columns that have exactly 2 dots.
3	MA.6.NSO.4.2 moderate	$(-9) \times (-5)$, $(-5) \times (-9)$, $(9) \times (5)$	Multiplying two negatives gives the same product as multiplying the two positives.
4	MA.6.AR.3.2 moderate	$18 \div 6$, $18/6$, $36 \div 12$	The unit rate is the total cost divided by the number of pounds.
5	MA.6.DP.1.5 moderate	15-19, 20-24	Count the values in each interval and compare.
6	MA.6.GR.2.2 moderate	63	The area of a parallelogram is base times height, using the height that is perpendicular to the base.
7	MA.6.GR.2.3 moderate	192	Volume is length times width times height, or the area of the base times the height.
8	MA.6.AR.3.1 moderate	7:9, 35:45, 42:54	Equivalent ratios simplify to the same pair of numbers in the same order.
9	MA.6.DP.1.6 moderate	Mean, Range, Maximum, Median	Adding a large value shifts the total and the greatest value, but it may leave the least value unchanged.
10	MA.6.NSO.2.1 easy	12	Multiply both numbers by the same power of 10 so the divisor is a whole number, then divide.
11	MA.6.DP.1.2 moderate	The mean is 14., The median is 14., The range is 13.	Check each statement by computing that measure from the data.
12	MA.6.NSO.4.1 moderate	$21 + (-15)$, $(-15) + 21$, $21 - (15)$	Addition can be done in either order, and subtracting a number equals adding its opposite.
13	MA.6.AR.1.4 moderate	$5x + 30$	Multiply the number outside the parentheses by each term inside.
14	MA.6.AR.2.3 moderate	$11x = 66$, $x \div 11 = 6/11$, $66 = 11x$	Substitute 6 for x and keep the equations that are true.
15	MA.6.NSO.1.1 easy	-3 1/4	-3 1/4 is more than -4 and less than -3, so it sits between them on the number line.
16	MA.6.DP.1.1 moderate	each student in the class	Asking about a whole group produces data that vary; asking about one person produces a single answer.

#	Benchmark	Answer	Explanation
17	MA.6.NSO.2.2 moderate	$\frac{2}{3} \times \frac{1}{2}$, $\frac{1}{2} \times \frac{2}{3}$, $\frac{2}{6}$	Multiplication of fractions can be done in either order, and the product of the numerators over the product of the denominators is the same value.
18	MA.6.NSO.1.2 easy	5 $\frac{3}{4}$	The opposite of a number is the same distance from 0 but on the other side.
19	MA.6.AR.1.3 moderate	103	Evaluate the exponent before multiplying, then subtract.
20	MA.6.NSO.1.4 easy	8	Take each absolute value first, then subtract.
21	MA.6.NSO.3.4 moderate	$2^2 \times 3 \times 7$, $2 \times 2 \times 3 \times 7$, 84×1	Each correct expression multiplies back to the original number.
22	MA.6.GR.1.3 moderate	4	Points with the same y-coordinate lie on a horizontal line, so subtract the x-coordinates.
23	MA.6.AR.2.4 moderate	$x + 1.5 = 9$, $x - 0.5 = 7$, $2x = 15$	Substitute 7.5 for x and keep the equations that are true.
24	MA.6.NSO.3.1 moderate	2, 3, 4, 6	A common factor divides both numbers with no remainder.
25	MA.6.NSO.3.5 moderate	$\frac{3}{20}$, 0.15, 15%	Equivalent forms name the same value as a fraction, a decimal and a percent.
26	MA.6.NSO.1.3 easy	3 $\frac{3}{4}$	The distance from 0 is the absolute value of the number.
27	MA.6.AR.1.1 moderate	$x - 4$	Translate each word phrase in order, keeping subtraction and division in the order they are written.
28	MA.6.GR.1.1 moderate	(-3, 4)	The first number tells how far to move right or left, and the second tells how far to move up or down.
29	MA.6.AR.3.5 moderate	8×8 , 64×1	Each cup is 8 fluid ounces, so multiply the number of cups by 8.
30	MA.6.AR.3.3 moderate	5:6, 10:12, 20:24	Multiplying both parts of a ratio by the same number keeps it equivalent; adding the same number does not.
31	MA.6.NSO.2.3 moderate	7.31×6 , 6×7.31 , $43.86 \div 1$	The total is the price times the number of tickets; order does not matter.
32	MA.6.GR.2.1 moderate	21	A triangle is half of a rectangle with the same base and height, so use one half times base times height.
33	MA.6.AR.1.2 moderate	$x \geq 36$	"At least" and "no more than" include the boundary value; "more than" and "under" do not.
34	MA.6.NSO.2.2 easy	6 $\frac{2}{3}$	Write the mixed number as a fraction, then multiply by the reciprocal of the divisor.
35	MA.6.NSO.3.2 moderate	$12(2 + 3)$, $24 + 36$, 12×5	Use the distributive property: the factored form must multiply back to the original sum.

#	Benchmark	Answer	Explanation
36	MA.6.DP.1.3 moderate	13, 14, 15, 17	The box itself covers the values from the lower quartile to the upper quartile.
37	MA.6.AR.3.4 moderate	$60\% \times 240$, $(60/100) \times 240$, 240×0.6	A percent of a number is that fraction of 100 multiplied by the number.
38	MA.6.AR.2.1 moderate	3	Substitute each value for x and check which one makes both sides equal.